M16C MCU

File: M16C MCU.kicad_sch

Motor drivers

File: motor drivers.kicad_sch

J102

RJ11 PIN6 via F100 (DNP) and trace cut +5V

TO_RJ11_PIN2_FROM_M16C_TXD1

RJ11_BOARD_NC_PIN5

TO_RJ11_PIN4_BOOT_TRIGGER_DTR

TO_RJ11_PIN_3_FROM_M16C_RXD1

RJ11 PIN1 is GROUND

GND

R244 420

R243 420

D1 LED

D2 LED

R241 2K7

LED

B

Q1 NPN

E C

R242 15K

LCD

File: lcd.kicad_sch

U401A

U401B

U401C

+5V

R168 100

IR1

VS

VO

GND

TSOP17TB1

R169 220

39K2 R170

R212C 1K

MCU_P2.5(INPC15)

IR_SIGNAL

GND

V+

R316 2K8

C408

2K2 R423

U401D

TP409 ${TP_SIGNAL_NAME}

R422 470K

2K2 R4

U302

OUT1 OUT2 OUT3 OUT4

V+ -IN1 +IN1 -IN2 +IN2 -IN3 +IN3 -IN4 +IN4 GND

LM339M

R???1 2K

TP411 ${TP_SIGNAL_NAME}

C407

3K9 R421

GND

V+

R2 2K8

Q403

C406

R210C 1K

MCU_P1.5

GND

+24V

+24V

Button boards

File: buttons.kicad_sch

J400

#Q400 P-MOSFET_SYMBOL

R?0 5K1

R408 5K1

T1 2N2222

R407 5K1

GND

Mow motor running if:
1G low
1C3 high
A high
B high

U400A

2G 1G B A 2C3 2C2 2C1 2C0 1C3 1C2 1C1 1C0 2Y 1Y

TP703 ${TP_SIGNAL_NAME}

Q401

10 R410

SHUNT0 SHUNT

R1 2K

GND

270 R404

R405 15K

Q402 N-MOSFETDPACK

15K R406

C410

R210D 1K

Mow enable A

MCU_P1.4

R210B 1K

Mow disable

MCU_P1.6

270 R403

R402 15K

R212B 1K

Mow enable B

MCU_P2.6

R210A 1K

270 R400

39K R401

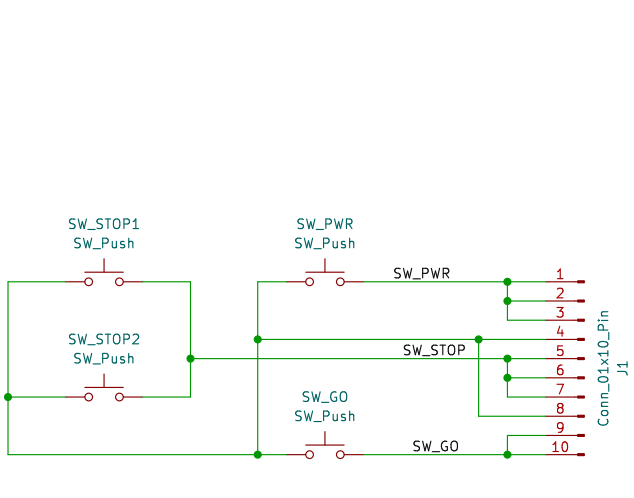
R???0B

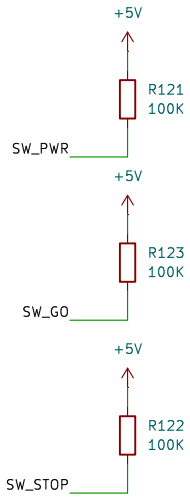
1K

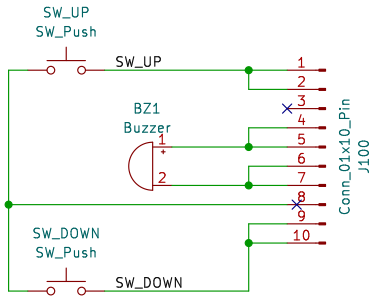
Motor PWM?

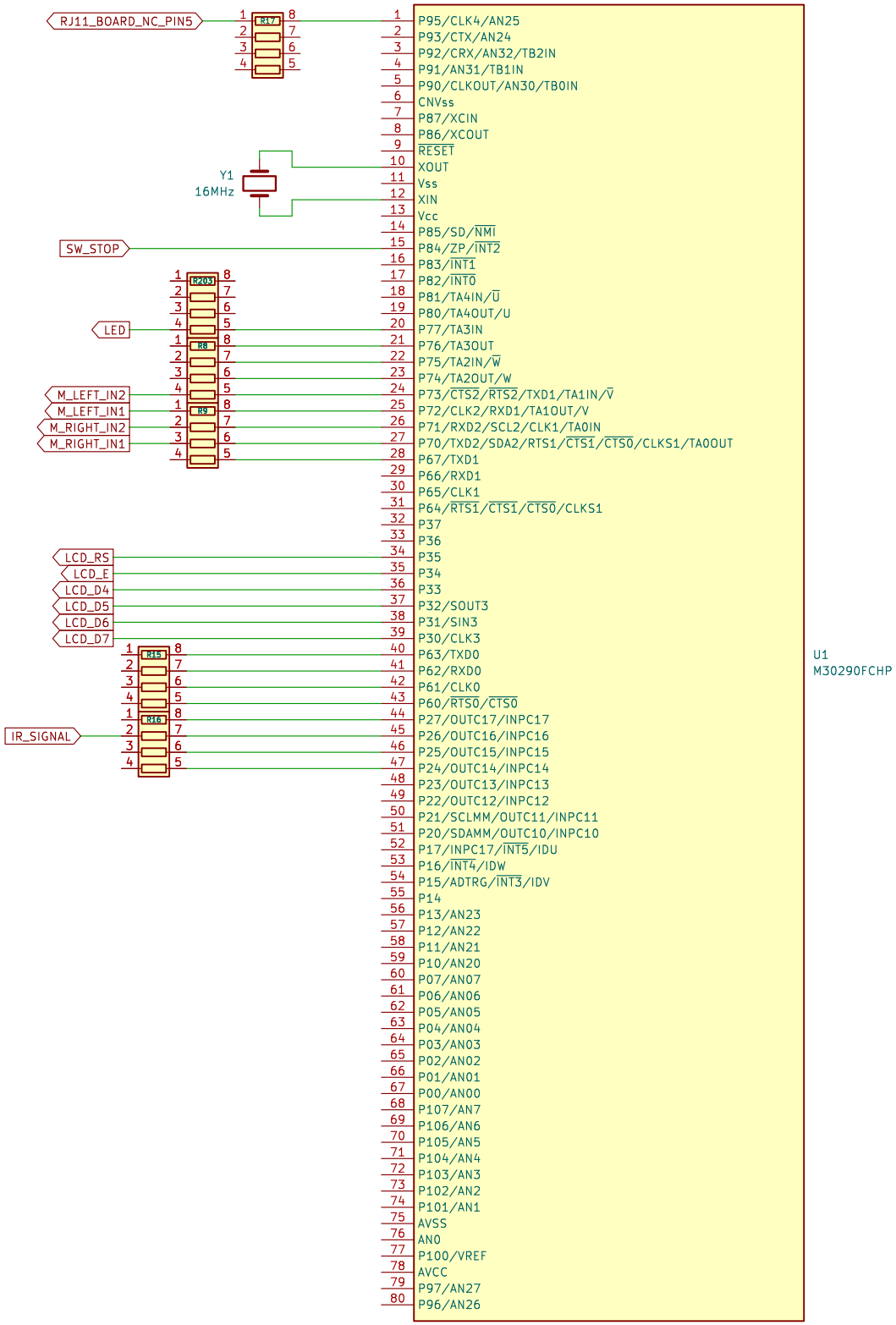
MCU_P?.?_TA2OUT

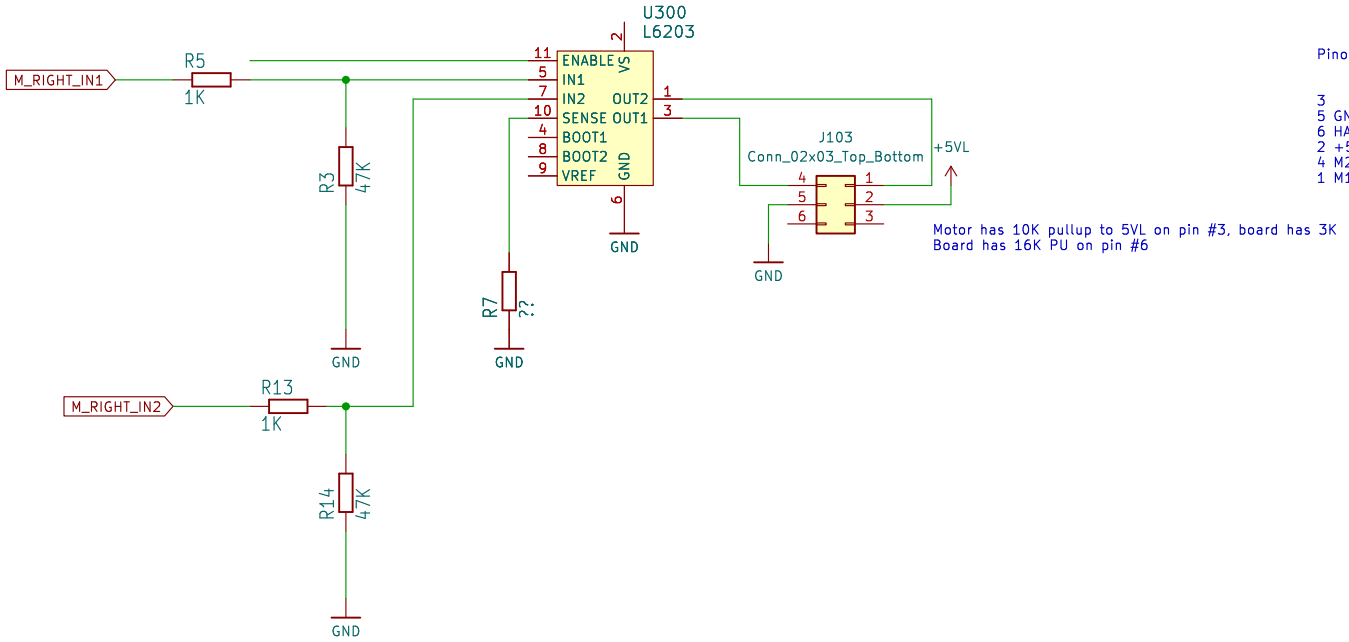
U400B

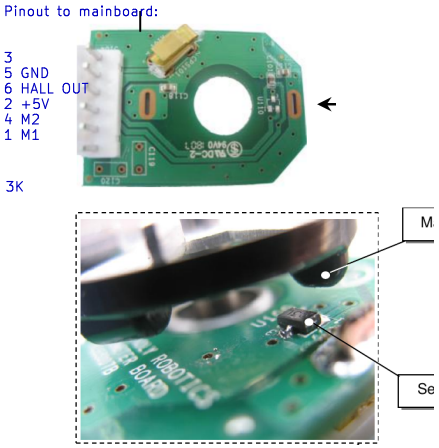

SW_STOP1
SW_Push
SW_STOP2
SW_Push
SW_PWR
SW_Push
SW_GO
SW_Push
SW_PWR
SW_STOP
SW_GO
1
2
3
4
5
6
7
8
9
10
Conn_01x10_Pin
J1

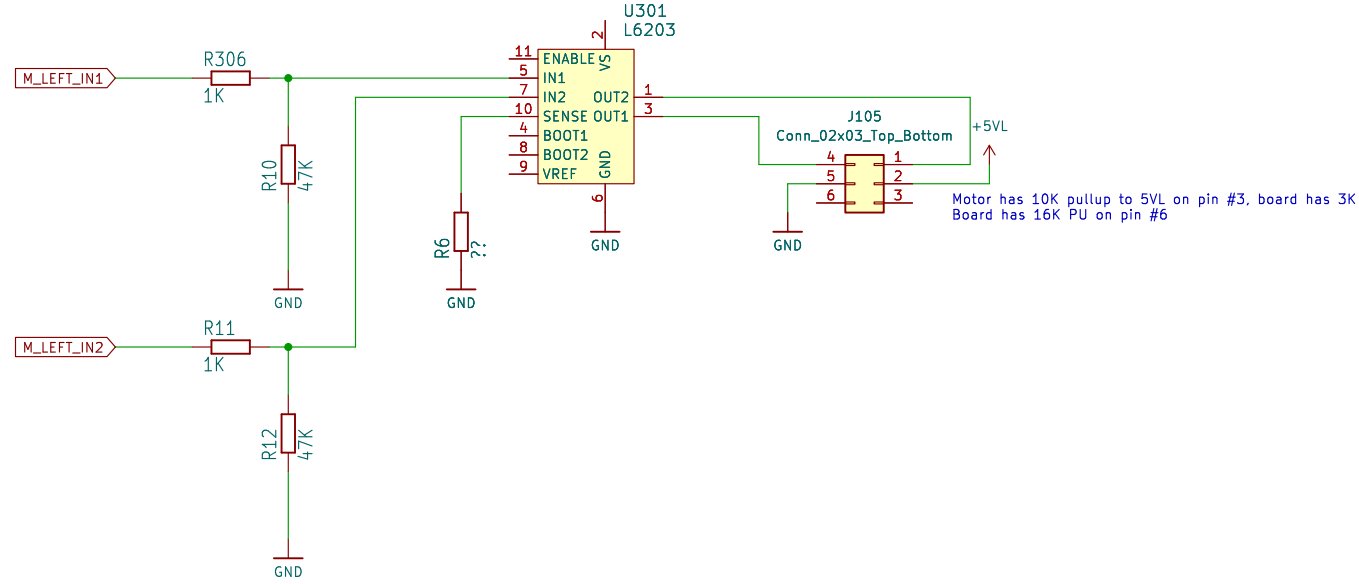

+5V
R121
100K
SW_PWR
+5V
R123
100K
SW_GO
+5V
R122
100K
SW_STOP

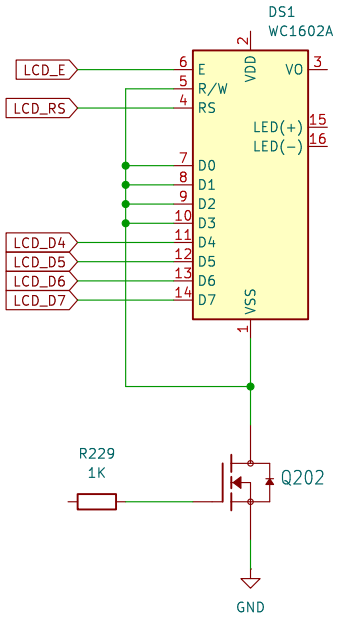

SW_UP
SW_Push
SW_UP
BZ1
Buzzer
1
2
SW_DOWN
SW_Push
SW_DOWN
1
2
3
4
5
6
7
8
9
10
Conn_01x10_Pin
J100


RJ11_BOARD_NC_PIN5
R17
Y1
16MHz
SW_STOP
RP203
LED
R8
M_LEFT_IN2
M_LEFT_IN1
M_RIGHT_IN2
M_RIGHT_IN1
R9
LCD_RS
LCD_E
LCD_D4
LCD_D5
LCD_D6
LCD_D7
R15
R16
IR_SIGNAL
P95/CLK4/AN25
P93/CTX/AN24
P92/CRX/AN32/TB2IN
P91/AN31/TB1IN
P90/CLKOUT/AN30/TB0IN
CNVss
P87/XCIN
P86/XCOUT
RESET
XOUT
Vss
XIN
Vcc
P85/SD/NMI
P84/ZP/INT2
P83/INT1
P82/INT0
P81/TA4IN/U
P80/TA4OUT/U
P77/TA3IN
P76/TA3OUT
P75/TA2IN/W
P74/TA2OUT/W
P73/CTS2/RTS2/TXD1/TA1IN/V
P72/CLK2/RXD1/TA1OUT/V
P71/RXD2/SCL2/CLK1/TA0IN
P70/TXD2/SDA2/RTS1/CTS1/CTS0/CLKS1/TA0OUT
P67/TXD1
P66/RXD1
P65/CLK1
P64/RTS1/CTS1/CTS0/CLKS1
P37
P36
P35
P34
P33
P32/SOUT3
P31/SIN3
P30/CLK3
P63/TXD0
P62/RXD0
P61/CLK0
P60/RTS0/CTS0
P27/OUTC17/INPC17
P26/OUTC16/INPC16
P25/OUTC15/INPC15
P24/OUTC14/INPC14
P23/OUTC13/INPC13
P22/OUTC12/INPC12
P21/SCLMM/OUTC11/INPC11
P20/SDAMM/OUTC10/INPC10
P17/INPC17/INT5/IDU
P16/INT4/IDW
P15/ADTRG/INT3/IDV
P14
P13/AN23
P12/AN22
P11/AN21
P10/AN20
P07/AN07
P06/AN06
P05/AN05
P04/AN04
P03/AN03
P02/AN02
P01/AN01
P00/AN00
P107/AN7
P106/AN6
P105/AN5
P104/AN4
P103/AN3
P102/AN2
P101/AN1
AVSS
AN0
P100/VREF
AVCC
P97/AN27
P96/AN26
U1
M30290FCHP


U300
L6203
M_RIGHT_IN1
R5
1K
R3
47K
GND
ENABLE
IN1
IN2
SENSE
BOOT1
BOOT2
VREF
VS
OUT2
OUT1
GND
J103
Conn_02x03_Top_Bottom
+5VL
R7
??
M_RIGHT_IN2
R13
1K
R14
47K
Motor has 10K pullup to 5VL on pin #3, board has 3K
Board has 16K PU on pin #6


Pinout to mainboard:
3
5 GND
6 HALL OUT
2 +5V
4 M2
1 M1


U301
L6203
M_LEFT_IN1
R306
1K
R10
47K
GND
ENABLE
IN1
IN2
SENSE
BOOT1
BOOT2
VREF
VS
OUT2
OUT1
GND
J105
Conn_02x03_Top_Bottom
+5VL
R6
??
M_LEFT_IN2
R11
1K
R12
47K
Motor has 10K pullup to 5VL on pin #3, board has 3K
Board has 16K PU on pin #6


DS1
WC1602A
LCD_E
LCD_RS
LCD_D4
LCD_D5
LCD_D6
LCD_D7
E
R/W
RS
D0
D1
D2
D3
D4
D5
D6
D7
VDD
VSS
VO
LED(+)
LED(-)
R229
1K
Q202
GND